



Computer-Assisted Surgery in Reverse Shoulder Arthroplasty: Early Experience

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Abstract

Background In the last decade, new technologies have been applied to shoulder arthroplasty. The aim of this work was to show that navigated RSA allows the surgeon to reach the planned version/inclination in all cases. In this article are shown preliminary data, advantages, disadvantages and limits of the technique.

Methods Eighteen computer-assisted reverse shoulder arthroplasty were performed. Preoperative glenoid version and inclination were evaluated with preoperative CT scan using Orthoblu[®] (Exactech, Gainesville, FL, USA) software, as well as baseplate type, planned glenoid component seating, planned postoperative version, planned postoperative inclination, intraoperative glenoid version/inclination, screw length and surgical time. A senior shoulder surgeon has analyzed the advantages, disadvantages and limitation of this kind of surgery.

Results Mean surgical time of the primary implants was 92 ± 12 min (min 75–max 110). Mean preoperative inclination was $+2.6^\circ \pm 6.4$, mean preoperative version was $-7.6^\circ \pm 8.4$. Mean planned postoperative inclination was $-2.7^\circ \pm 2.3$, mean planned postoperative version was $-1.6^\circ \pm 2.9$ and mean planned glenoid seating was $89\% \pm 8\%$. Planned settings were reached in all cases during surgery. Baseplate implanted were in nine cases 8° posterior augmented, in six cases standard and in three cases 10° superior augmented. Mean screw length was $33.5 \text{ mm} \pm 4.2 \text{ mm}$. No GPS system failure has been recorded. One coracoid fracture occurred during the first case.

Discussion Intraoperative navigation system is a reliable and user-friendly technology that allows the surgeon to reach planned glenoid positioning during surgery. Furthermore, this technology will allow the surgeon to compare clinical outcomes to component positioning data. The lack of humeral implant navigation is the main limit of this technique.

Keywords Computer-assisted RSA · Reverse shoulder arthroplasty · Navigation · Navigated reverse shoulder · Arthroplasty · Computer-assisted surgery reverse shoulder arthroplasty · Shoulder replacement

Introduction

Reverse shoulder arthroplasty (RSA) was introduced in 1985 by Grammont [1] to treat gleno-humeral arthritis. This prosthesis allowed surgeons to treat eccentric gleno-humeral arthritis with rotator cuff degeneration showing excellent functional outcomes. Despite this, several complications have been recorded such as scapular notching that affected two-third of Grammont RSA [2] and a relatively high rate of glenoid loosening [3]. In these 35 years, several improvements in prosthetic design and RSA biomechanic knowledge

have been reached [4], and several studies on glenoid positioning have been performed, thus reducing complication rates and glenoid mobilization [5].

As long as it is known, glenoid positioning should be flush to the inferior rim of the glenoid to decrease scapular notching and reduce the impingement [4]. Avoiding superior tilt of the glenoid component reduces the risk of scapular notching and glenoid component loosening and improves the range of movement [6]. Inferior tilting of the glenoid is still controversial [7]: an inferior tilt between 10° and 15° can reduce dislocation rate and decrease the total micromotion over the full range of abduction, with a more even distribution of compressive forces beneath the baseplate [8]. However, tilting the implant requires the removal of cortical bone at the inferior pole, which has been addressed to increase the risk of glenoid loosening

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[9]. For these reasons, a neutral tilt positioning is generally used as goal [10].

Regarding glenoid version, most surgeons will attempt to reduce glenoid retroversion to a neutral version, or at least 10° to 15° as clinical and biomechanical data support increased risk for glenoid component loosening if the glenoid is placed in higher degrees of retroversion [10].

During surgery, it is not easy to understand the morphology and orientation of the glenoid and how to reach the previously described goals. Recently, glenoid inclination and version have been investigated with preoperative planning software using 3D CT scan which have shown more reliability than 2D CT scan measurements [11–13]. The advent of CT-based preoperative planning software has perhaps revolutionized surgeons mentality: several authors have shown how the use of preoperative planning software improves the capability of the surgeons to achieve the desired positioning of the glenoid component in RSA [10, 14, 15]. In some shoulders, it is difficult to reach the right balance between glenoid seating and suitable positioning: preoperative planning helps the surgeon to find this right balance. Unfortunately, they can help surgeon only during the preoperative phase, without granting a real-time assistance during the intraoperative phase.

In the last years, RSA-affirmed efficacy in glenohumeral arthritis treatment fostered the widening of its indications even to younger patients who have greater functional expectations. Due to the widening of the indications, surgeons showed increasing interest in the application of new technologies in RSA world such as patient-specific implants, preoperative planning software and computer-assisted intraoperative navigation [16]. These technologies aim to reach better accuracy in implant positioning that could result in better clinical outcomes, even if this has not yet been proved. Further steps to help surgeons intraoperatively are patient-specific instruments (PSI) and computer-assisted navigation system: first experiences showed promising results in improving the accuracy of glenoid component positioning [17, 18]. Only about 30 articles have been published on computer-assisted RSA: a few little case series have been published, while some studies focused on glenoid positioning accuracy or screw positioning. In a recent meta-analysis of Burns et al. [20], only 2 articles were considered for computer-assisted surgery in shoulder [17, 18]: in these 2 articles were collected 17 patients. Therefore, very little is known regarding this technology. The aim of this work is to show that computer-assisted RSA helps the surgeon reach the preoperative plan during surgery. The secondary aim of the work is to determine the amount of increase in surgical time and screw length.

Materials and Methods

This is a case series report of all computer-assisted primary RSA performed at our department since October 2018 to November 2019. Guided Personalized Surgery (GPS) navigation system (Exactech, Gainesville, FL, USA) was used to assist the implant of Equinox reverse shoulder prosthesis (Exactech, Naples, FL, USA) in all cases. Inclusion criterion was patients undergoing GPS-assisted primary RSA; no exclusion criterion was applied. Patients were selected as a sample of convenience. Eighteen computer-assisted RSA were included in the study. Four patients were male, 14 were female. Mean age at the time of surgery was 75 years (min 62–max 87). Indications were seven rotator cuff tear arthropathy, four concentric arthritis, three rheumatoid arthritis, two post-traumatic arthritis, one locked posterior luxation, one four-part proximal humeral fracture and one RSA two-stage revision. Walch classification was A1 in two cases, A2 in two cases, B1 in six cases, B2 in four cases and B3 in four cases. Associate procedures were performed in two cases: one posterior bone graft in a severe bone loss and one plate removal in a post-traumatic case. All data available in clinical records were collected: preoperative glenoid version and inclination evaluated with preoperative CT scan using Orthoblu[®] (Exactech, Gainesville, FL, USA) software, baseplate type, planned glenoid component seating, planned postoperative version and planned postoperative inclination. Intraoperative glenoid version and inclination was measured with GPS, while screw length and surgical time were assessed during surgery. Data were successively analyzed by a shoulder surgeon to find the main advantages, disadvantages and limitations of this kind of surgery. Mean surgical time of primary implants was then compared with the last 15 primary implants performed without navigation. For learning curve considerations, the mean time of the first eight primary implants without any other associate surgery has been compared with the other primary implants without any other associated surgery with the Student's T test to search a statistically significant difference. The cut-off of eight surgeries was taken from a previous study of Wang [19]. Finally, the mean screw length has been compared with the one in the last 15 Equinox RSA implanted with the traditional method with the Student's T test.

Pre-operative Phase

Once surgical indication for RSA was given during the outpatient visit, shoulder CT scan was uploaded with Orthoblu software (Exactech, Gainesville, FL, USA) for

GPS reconstruction. During the upload phase, the software provided automatic data anonymization. Surgery data were then selected at least 10 days after the upload because the production of a virtual 3D scapular model elaborated by Orthoblu engineers takes 10 days. Emergency procedures allowed a 7-day reconstruction. An immediate reconstruction was also available (only for preoperative planning use) with scapular landmarks appointed by the surgeon.

Once the GPS reconstruction became available, surgeon selected the type of baseplate and chose the final position of the implant, which then have to be exported to USB key and then to GPS hardware.

Surgical Technique

Beach chair position was used. Computer interface with a touchscreen monitor was included in the surgical field, setting it with its own support-arm. Deltoid-pectoral approach was performed, widened proximally of about 1.5 cm to allow a better view of the coracoid process. Flake osteotomy of the lesser tuberosity was performed. Humeral osteotomy and broaching were performed with the standard toolkit. Before glenoid exposition, the superior face of the coracoid process was prepared to coracoid tracker positioning. Two screws were used: the first longer screw was positioned towards the coracoid base while the shorter screw was positioned in the distal portion. After that, glenoid exposition with soft tissue removal was carefully obtained and then the landmarks acquisition started. At the end of the acquisition process, a check of the landmark acquisition's accuracy was performed by the software: if a lack of accuracy was found, landmarks had to be acquired again (Fig. 1). Pilot hole in the glenoid was performed using a proper drill, with a tracker on the handle communicating with the monitor, following the navigation system indications. Glenoid reaming (Fig. 2) and cage hole were subsequently performed with proper toolkit, always following navigation guide. Glenoid component was

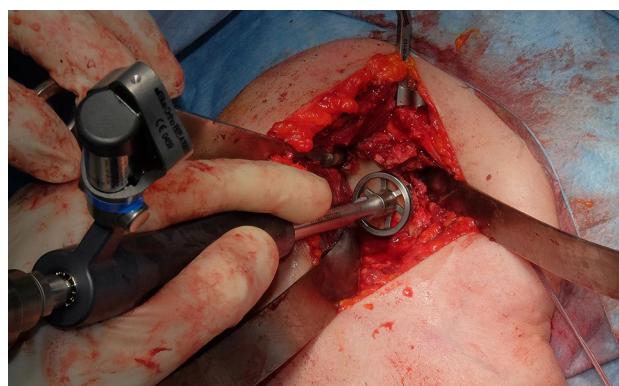


Fig. 2 Glenoid reaming with a tracker on reamer's handle allow navigation assistance

then implanted, without navigation aid. Equinoxe system provides six compression screw holes with 30° angular variability. Usually three or four screws have been used (one in the superior hole and two or three in the inferior holes). Screw positioning was guided by navigation system (Fig. 3); length of the screws was measured with a color system applied on the drillbit. Coracoid tracker was removed. Glenosphere was applied as usual. At last, humeral component was implanted with the traditional toolkit, as well as reduction and transosseous suture of the lesser tuberosity.

Results

Eighteen computer-assisted primary RSAs were included in the study. Mean surgical time of the primary implants without any other associated surgery was 92 ± 12 min (min 75—max 110), which compared with the mean surgical time of the last 15 primary implants performed at our institute without navigation (75 ± 12 min, min 52—max 90) showed a statistically significant difference ($p=0.001$). The first eight

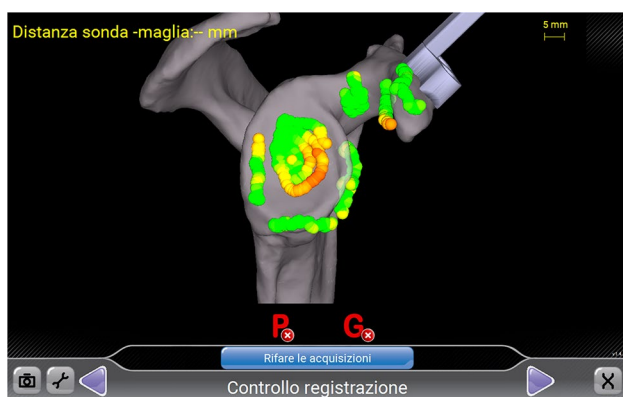


Fig. 1 Screenshot of the landmark check after acquisition: if red points are present, it is necessary to acquire landmarks again

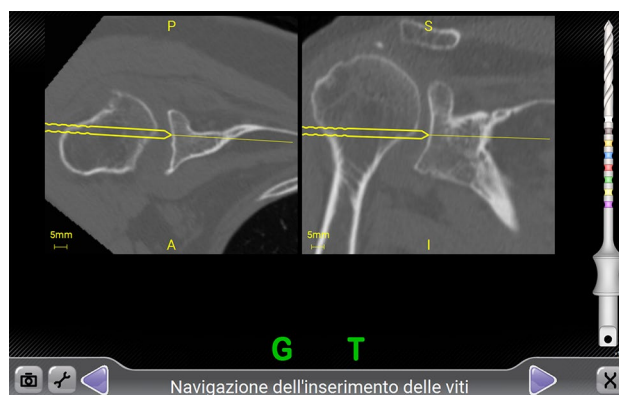


Fig. 3 Screenshot of the navigator monitor during screw positioning

cases showed a mean surgical time of 97 ± 6 min against the 85 ± 13 min of the other cases, with a significant difference between the two groups ($p=0.035$). Mean preoperative inclination was $+2.6^\circ \pm 6.4$ (min -7° , max 18°), while mean preoperative version was $-7.6^\circ \pm 8.4$ (min -27° , max $+2^\circ$). Mean planned postoperative inclination was $-2.7^\circ \pm 2.3$ (min -6° , max 0°), mean planned postoperative version was $-1.6^\circ \pm 2.9$ (min -7° , max $+3^\circ$) and mean planned glenoid seating was $89\% \pm 8\%$. In all cases, the preparation of the implant site has been done as planned, with no deviation from the planned version/inclination. Baseplate implanted were in nine cases 8° posterior augmented, in six cases standard and in three cases 10° superior augmented. Mean screw length was 33.5 ± 4.2 mm. Mean screw length of the last 15 cases implanted with the traditional method was 29.1 ± 3.9 mm, with a significant difference in the navigated group ($p < 0.05$). No GPS system failure has been recorded: all surgeries were effectuated without technical problem in all cases. One coracoid fracture occurred during the first case.

Discussion

Considering the novelty of the technique used, this is actually the second widest sample series on computer-assisted shoulder prosthesis implant. In this article, the authors give some tips and tricks to avoid mistakes during surgery and show early data on computer-assisted surgery.

First of all, GPS navigation system has shown no failure, granting in all cases an excellent precision in gain of the previously planned glenoid orientation during surgery. In all the cases, the surgeon has reached a preparation of the site as planned. The planned seating of $89\% \pm 8\%$, which we can assume has been approximately gained with navigation aid, is an excellent goal. Especially in the most difficult cases, it is not easy to reach a good glenoid seating using a standard toolkit: surgeons often fail to evaluate the right glenoid orientation [20]. The navigation toolkit allowed a very accurate preparation of the site of implant [21], and its use was comfortable even in most difficult cases: in this series, the surgeon has been able to reach the desired planned glenoid positioning in each case.

A younger surgeon can benefit the most from intraoperative navigation increasing his glenoid “awareness” and reducing his RSA learning curve and avoiding mistakes that in first cases could occur.

The better awareness of the surgeon about the glenoid position in the space given by pre-operative planning software leads to an increase in glenoid augmentation use: 12 out of 18 implants were superiorly or posteriorly augmented, thus confirming literature data [11]. However, surgeons should keep attention to carefully evaluate the

preoperative planning because preoperative software version can under- or overestimate inclination leading to an improper component planning [22].

Intraoperative navigation system is a helpful tool to assess intraoperatively the glenoid orientation, prepare the site of implant as previously planned and to control the direction and the length of the screws. No failure of the navigation system has been recorded, while in the literature, one (2.7–4.2%) case in each study failed [14, 19]; the navigation system is, therefore, strongly reliable. This technology is quite user friendly, with a relatively short learning curve; a previous study claims that eight surgeries performed are enough to acquire navigation proficiency [19], in this case series, there is a statistically significant difference in the mean surgical time between the 1st eight primary implants and the other seven primary implants, confirming previously published data. Literature showed an increase in surgical time of approximately 6 min [14]; in this series, the mean surgical time of navigated implants differs by 17 min from the traditional implants, but if you consider only the mean surgical times of the last seven implants of the navigated series, the difference is about 10 min.

The operative room setting required a little more attention, a little trick for patient positioning is trying to keep the head support as proximal as possible to avoid the interposition of the support between the coracoid tracker and the monitor. Furthermore, sometimes, scalytic lights can interfere with the optical tracking system. In this case, the tracker will disappear from the screen; therefore, it is necessary to move the scalytic light to avoid interferences. Considering the surgical technique, when fixing the coracoid tracker, the tracker should be kept as proximal as possible to reduce coracoid fractures which can occur. One case of coracoid fracture has been recorded in our series, according to the literature [14, 19]. During all surgeries, tracker must not be in conflict with soft tissue or skin; otherwise, the navigation system will not be accurate. Removing soft tissue from glenoid before starting landmark registration is necessary to obtain a good exposure of the glenoid neck inferiorly and anteriorly without any osteophyte removal; otherwise, accuracy of the registration will be compromised. Glenoid exposure takes a little more time than the traditional method to avoid landmark recording mistakes.

The navigated screw fixation allowed the surgeon to avoid cage perforation and to reach a better screw purchasing and positioning [23, 24]. In this series, an increase in screw length has been recorded, with a statistically significant difference with respect to the traditional method. Controlled screw direction and length is a unique feature of intraoperative navigation, and it is a real improvement in surgeon's performance because screws provide the majority of initial glenoid fixation in RSA [23].

One limit of the GPS navigation system was the lack of navigated control of final implant positioning: the rotation of the component is not navigated; therefore, a wrong rotation during the implant could cause a wrong positioning in an elliptical baseplate with an eccentric cage. Knowing the final implant positioning would be a paramount feature to study the best glenoid positioning, allowing a real investigation on how the glenoid position influences the outcomes.

The main limit of this technology is that the humeral osteotomy and implant positioning were completely neglected by the navigation system; even if correct glenoid positioning and fixation could be a key point for implant survival and postoperative outcome, the humeral component plays a paramount role in stability, deltoid efficiency and joint loading [1, 25–27]. Intraoperative navigation system can, therefore, be the instrument which will allow researchers to find the right “combined” version and inclination of the glenoid/humeral system which is the aim of RSA. Soft tissue tension, most of all deltoid tension, should be another issue to investigate; for this reason, it is important to have a complete view of the replaced joint, not only of the glenoid component.

There were several limits to the study: the sample was small, population showed high variability in diagnosis and no clinical outcomes have been reported. However, this study is only meant to discuss how navigation can improve surgery in the present and in the future.

Conclusion

Intraoperative navigation system is a reliable and user-friendly technology that allows the surgeon to implant the glenoid component with the desired pre-planned positioning with a better accuracy in glenoid and screw positioning. Furthermore, this technology will allow the surgeon to compare clinical outcomes to component positioning data. The lack of humeral implant navigation is the main limit of this technique.

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Data Availability The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Compliance with Ethical Standards

Conflict of interest The authors declare that they have no competing interests.

Ethical standard statement This study was approved by Ethics Committee of Policlinico di Modena—University of Modena and Reggio Emilia, protocol no. 772/2019.

Informed consent Rules for informed consent were observed in accordance with the Ethical Standards of the 1964 Declaration of Helsinki as revised in 2000.

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